

ACTIVITY-05

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Q1. Geospatial Data Visualization

Program:

```
library(ggplot2)
```

```
data <- data.frame(
```

```
  City=c("Chennai","Bangalore","Hyderabad","Mumbai","Delhi","Pune","Kolkata","Ahmedabad","Jaipur","Kochi"),
```

```
  Latitude=c(13.08,12.97,17.38,19.07,28.61,18.52,22.57,23.02,26.91,9.93),
```

```
  Longitude=c(80.27,77.59,78.48,72.87,77.21,73.85,88.36,72.57,75.79,76.26),
```

```
  Population=c(11,13,10,20,32,7,15,8,4,2)
```

```
)
```

```
ggplot(data,aes(x=Longitude,y=Latitude,size=Population,color=Population,label=City))+
```

```
  geom_point(alpha=0.7)+
```

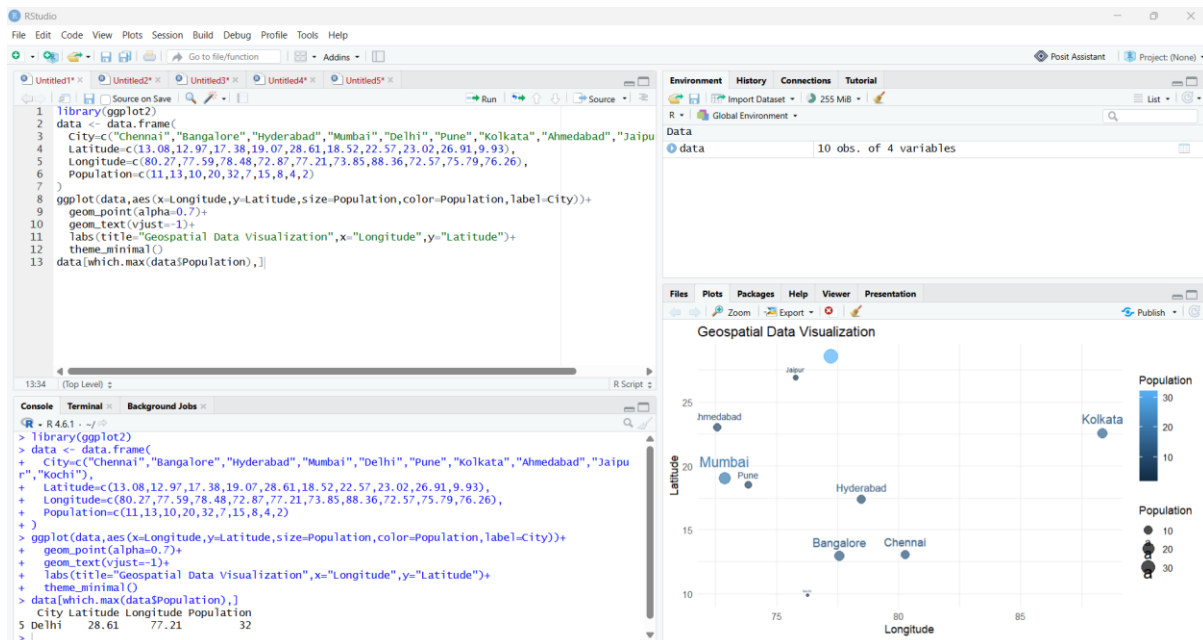
```
  geom_text(vjust=-1)+
```

```
  labs(title="Geospatial Data Visualization",x="Longitude",y="Latitude")+
```

```
  theme_minimal()
```

```
data[which.max(data$Population),]
```

OUTPUT:



Q2. Network Visualization

PROGRAM:

```
library(igraph)
```

```
edges<-data.frame(
```

```
  From=c("A","A","B","C","C","D","E","F","G","H"),
```

```
  To=c("B","C","D","D","E","E","F","G","H","A")
```

```
)
```

```
g<-graph_from_data_frame(edges,directed=FALSE)
```

```
plot(g,
```

```
  vertex.color="skyblue",
```

```
  vertex.size=30,
```

```
  vertex.label.color="black")
```

```
degree(g)
```

```
names(which.max(degree(g)))
```

```
plot(g,
```

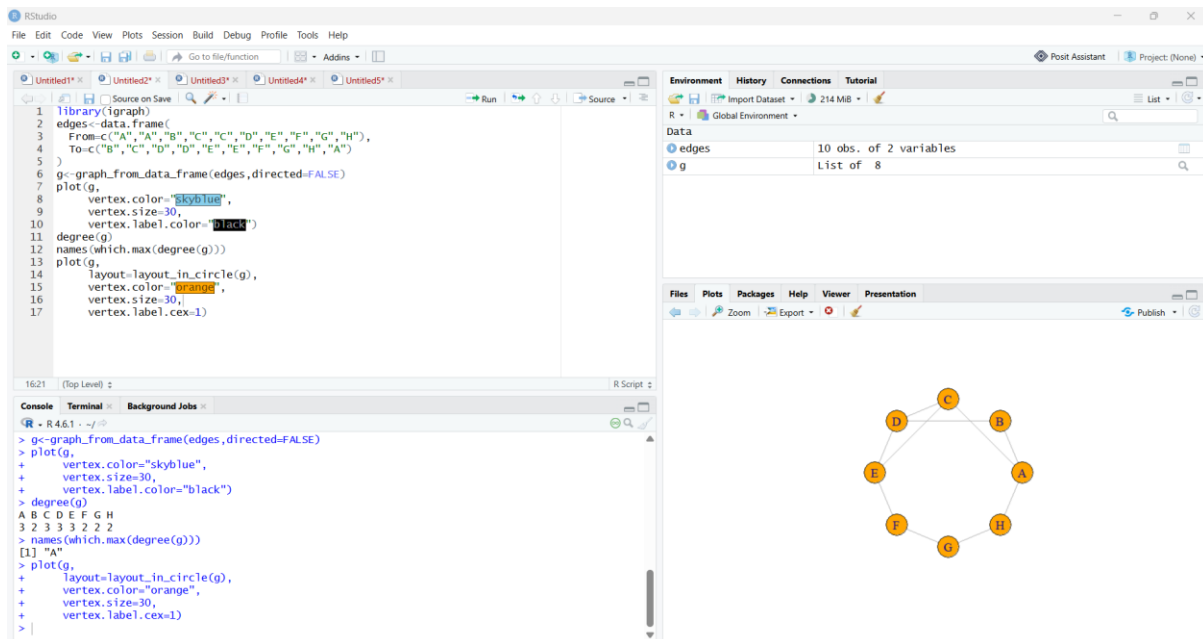
```
  layout=layout_in_circle(g),
```

```
  vertex.color="orange",
```

```
  vertex.size=30,
```

```
vertex.label.cex=1)
```

OUTPUT:



Q3. Temporal Data Representation

PROGRAM:

```
month<-c("Jan","Feb","Mar","Apr","May","Jun","Jul","Aug","Sep","Oct","Nov","Dec")
```

```
sales<-c(120,135,150,170,190,220,210,230,250,270,290,320)
```

```
plot(1:12,sales,
```

```
  type="o",
```

```
  pch=19,
```

```
  xaxt="n",
```

```
  xlab="Month",
```

```
  ylab="Sales (Lakhs)",
```

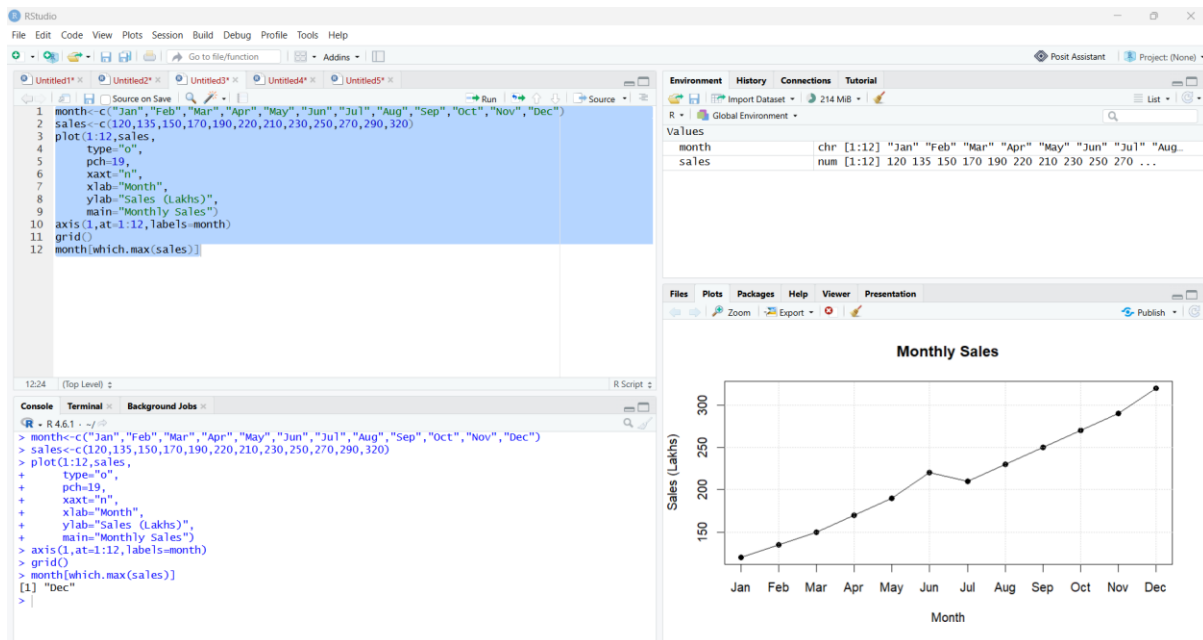
```
  main="Monthly Sales")
```

```
axis(1,at=1:12,labels=month)
```

```
grid()
```

```
month[which.max(sales)]
```

OUTPUT:



Q4. Multivariate Data Visualization

Program:

```

student<-data.frame(
Student=c("A","B","C","D","E","F","G","H","I","J"),
Marks=c(60,68,74,79,82,85,88,91,94,97),
Attendance=c(75,80,84,88,90,92,94,95,97,99),
StudyHours=c(2,3,4,5,5,6,6,7,8,9),
Projects=c(1,2,2,3,3,4,4,5,5,6)
)

```

```

symbols(student$Marks,
student$Attendance,
circles=student$StudyHours,
inches=0.3,
bg="blue",
fg="black",
xlab="Marks",

```

```
ylab="Attendance",
```

```
main="Bubble Plot")
```

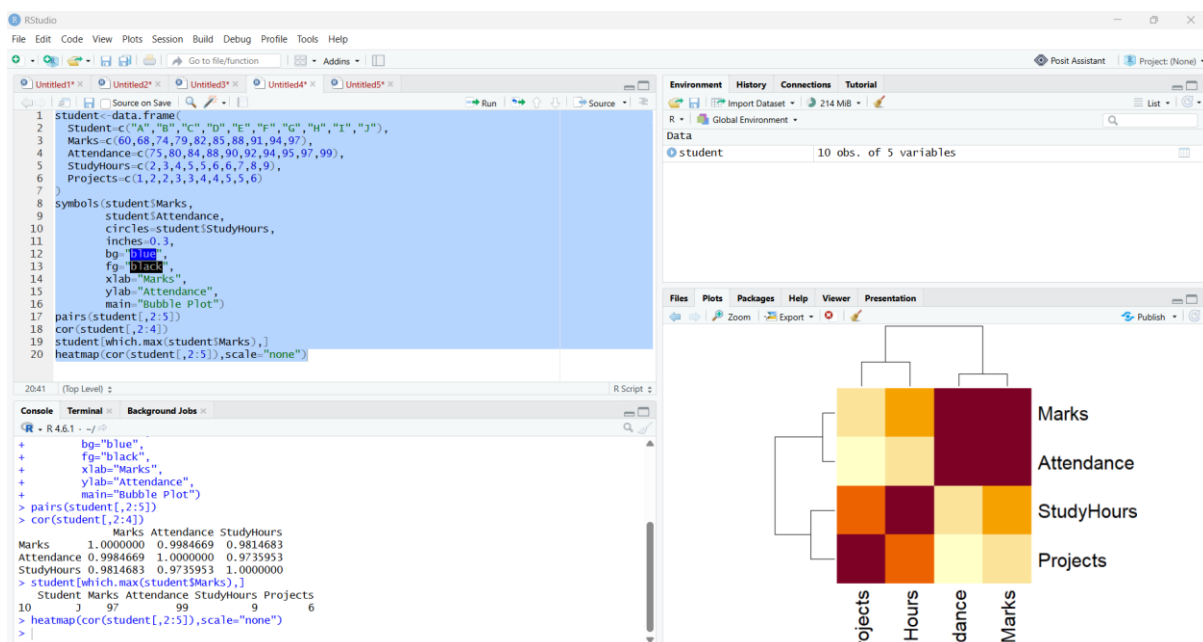
```
pairs(student[,2:5])
```

```
cor(student[,2:4])
```

```
student[which.max(student$Marks),]
```

```
heatmap(cor(student[,2:5]),scale="none")
```

Output:



Q5. Integrated Data Visualization Exercise

Program:

```
data<-data.frame(  
Year=2016:2025,
```

```
Sales=c(150,170,200,240,260,300,340,390,430,480),
```

```
Profit=c(20,25,32,40,42,55,65,72,82,95),
```

```
Customers=c(1200,1500,1800,2200,2400,2800,3300,3800,4200,4700),  
Branches=c(5,6,7,8,8,10,11,12,13,15)  
)
```

```
plot(data$Year,  
data$Sales,  
type="o",  
pch=19,  
xlab="Year",  
ylab="Sales",  
main="Yearly Sales")
```

```
symbols(data$Sales,  
data$Profit,  
circles=data$Customers,  
inches=0.3,  
bg="green",  
fg="black",  
xlab="Sales",  
ylab="Profit",  
main="Bubble Plot")
```

```
cor(data[,2:5])
```

```
pairs(data[,2:5])
```

```
heatmap(cor(data[,2:5]),scale="none")
```

Output:

